Phase 2: Extended Literature Review and Research Methodology Development

Adaptive Network Intrusion Detection and Early Warning through Game-Theoretic Scenario Generation and Temporal Network Analysis

1. Research focus and continuity with Phase 1

Phase 1 established that novelty in network intrusion detection cannot be inferred from the use of graph neural networks or adversarial training alone. Deepanramkumar and Sharmila [P1] demonstrate benign-only graph-temporal anomaly reconstruction; Bala et al. [P2] show robust temporal flow-graph classification under controlled feature perturbations; and Afrifah et al. [P3] demonstrate stage-aware network classification but explicitly cannot validate pre-impact prediction because their held-out partitions lack positive precursor-to-impact sequences. The most pertinent research opening is therefore *not* generic "zero-day detection": it is whether a network telemetry testbed can produce independently timed, reproducible multi-stage behaviour and whether a bounded strategic scenario policy generates more informative early-warning data than fixed or random policies.

Our question is:

> *Can game-theoretic selection of controlled adversarial network scenarios improve causal detection and pre-milestone early warning for previously unseen multi-stage communication behaviours, compared with fixed and uniformly random scenario selection under equal budgets and held-out network conditions?*

The project is primarily *Computer Networks* research: measured TCP/UDP flows, protocol metadata, communication-graph structure, temporal evolution, inter-segment movement and traffic transfer. Cybersecurity provides bounded incident scenarios; machine learning provides detection models. Detection and early warning are required. Operational response and future-stage prediction beyond a measured warning score are optional. No deployed malware, authentic zero-day exploitation, autonomous exploit discovery or Kubernetes cluster is proposed.

2. Extended literature review: ten additional studies

The ten additional 2026 papers below are distinct from Phase 1's three focal studies and meet the supplied 2025–2026 period. They cover four complementary streams. **Forecasting:** Nayeri and Rezvani [1] model evolving attacker–victim alert graphs and predict future links and attack types; this is a stronger temporal forecasting comparison than an ordinary flow classifier. **Strategic/adversarial adaptation:** Ioannou et al. [2] co-evolve RL attackers and adaptive detection; Ding et al. [3] use a cost-sensitive two-player minimax objective, showing that game-theoretic learning already exists. **Generalisation and validity:** Eren et al. [4] study distribution shift across distinct IDS corpora, Mishra et al. [5] study feature harmonisation and statistical validation, and Nune et al. [6] show that leakage controls and even graph node identification can reverse apparent model rankings. **Network model dynamics:** Wang et al. [7] study resource-aware dynamic inference, Yao and Koirala [8] study temporal graphs under faults and adversarial effects, Javed et al. [9] study continual temporal graph learning, and Suharsono et al. [10] examine coordinated multi-stage enterprise activity.

The articles are methodological comparators, *not* evidence that any common benchmark simultaneously validates early warning, hostile scenario policy and unseen-topology transfer. Reported results below belong to their respective authors and cannot be directly pooled because tasks, labels, splits, class ratios and network observability differ. Journal quartiles are category/year specific (2025 SJR or JCR used as noted later). The official task permits Q1–Q2 journals; Suharsono et al. [10] qualifies by 2025 SJR Q2 although a 2025 JCR source lists Q3. No unsupported novelty or deployment-performance claim is made.

Cross-paper comparison (2025–2026) – studies 1–5

#	Paper / Venue	Year	Dataset / Source	Preprocessing / Representation	Models / Techniques	Evaluation Metrics	Reported Finding	Limitations / Research Gap
1	TGNE-TA; *JNCA*	2026	Real-world attacker–victim alert streams; authors cite released archive	Alert-pair time windows, dynamic bipartite graph, local and global time encoding	TGN; TGNE edge labels; Transformer attention aggregation	Next victim/link; multiclass alert type accuracy; eight comparison networks	Multiclass predictive accuracy above 90% under tested data	Dependent on earlier alerts and historical attacking entities; not raw-flow warning before a logged impact milestone.
2	RL active NIDS; *JISA*	2026	DReLAB; additional CICIDS2017 tests	Fixed safe splits; scaled 39-feature flows; adversarial and target distributions	A2C, PPO, DDQN, SAC attackers; adaptive NIDS; WGAN-GP screening	Attack success; malicious-class F1; adversarial detection rate; FPR; runtime/energy	Static malicious F1 collapsed under tested RL attacker; WGAN-GP ADR 96.9%; best hardened F1 1.0	Controlled feature-level attack game; data on request; not pre-impact live network episodes.
3	GT-CSAT; *Applied Sci.*	2026	CVD; CICIDS2017; UNSW-NB15	Attack-conditioned perturbation; cost-weighted samples and shared detector inputs	WGAN-GP + game-theoretic minimax cost-sensitive training	Robust accuracy; clean accuracy; perturbation-family comparisons	Mean robust accuracy 94.3%; clean accuracy 97.1% across specified setting	Synthetic evasion on static data; code release claimed without a clear independently verified source URL in provided PDF.
4	IoT distributional drift; *Electronics*	2026	BoT-IoT; ToN-IoT; UNSW-NB15	Harmonised labels/features, distribution-preserving sampling, feature propagation and correlation reduction	Distribution-shift diagnosis and cross-dataset ML benchmarking	Cross-domain classification and feature-distribution comparisons	Demonstrates degradation on unseen datasets despite strong within-source scores	Observational dataset drift; no controlled attacker–defender policy or milestone-labeled lab.
5	CrossHIDS; *PLOS ONE*	2026	NSL-KDD; CICIDS2017	Manual feature alignment; train-only preparation; hash-chained experiment log	Supervised, one-class and ensemble models	F1; accuracy; Wilcoxon, McNemar and DeLong analyses	Best RF: 98% accuracy and 97% F1 in reported harmonised task	Manual alignment is brittle; no prospective warning timeline or game-controlled data generation.

Comparison note. JNCA = Journal of Network and Computer Applications; JISA = Journal of Information Security and Applications; CVD = Cloud Vulnerabilities Dataset. Numerical values are tied to each source's specific protocol. In particular, the perfect post-hardening score in [2] does not demonstrate perfect real-world detection.

Cross-paper comparison (2025–2026) – studies 6–10

#	Paper / Venue	Year	Dataset / Source	Preprocessing / Representation	Models / Techniques	Evaluation Metrics	Reported Finding	Limitations / Research Gap
6	STAR-IDS; *Scientific Reports*	2026	NF-CSE-CIC-IDS2018-v3; NF-ToN-IoT-v2	Chronological tenflow graphs; real endpoint identities; blocked partitions	Two-level attention; prototype memory; MC-dropout; MLP/GNN baselines	Macro-F1; accuracy; uncertainty AUC; paired seeds and Holm correction	86.41% macro-F1 on NF-CSE; no statistically significant gain; MLP wins on NF-ToN-IoT	Shows leakage/graph sensitivity; no independent ground-truth multi-stage alert lead time.
7	IDNet; *Computer Networks*	2026	Heterogeneous NetFlow and CIC benchmark corpora	Traffic-vector-to-image instantiation, structural similarity constraints	Adversarial training; adaptive residual detector with lightweight policy router	Accuracy; inference speed and efficiency; unknown-class sensitivity	Reduced inference cost while preserving competitive accuracy; platform-speed gains reported	Policy routing selects model computation, not attacker–defender network scenario strategy.
8	DTGNN-FIDS; *Cluster Computing*	2026	Edge-IIoTset; ToN-IoT; IoT-23 with mobility/fault injection	Time-varying host/edge graphs, event-triggered telemetry, fault settings	Temporal GNN + adversarial regularisation and edge/cloud allocation	Macro-F1; latency; energy; bandwidth overhead under failures	Macro-F1 $\geq$ 92.4% for stated faults $\leq$ 20%; telemetry bandwidth down about 68%	Cyber-physical, injected-fault context; no budget-matched game for early lead time.
9	HECATE-IDS; *Pervasive and Mobile Computing*	2026	ToN-IoT; BoT-IoT; MQTT-IoT; CICIoT2023	Temporal interaction graphs; sequential tasks and non-IID client partitions	Continual temporal graph model; explainability; federated evaluation	F1; accuracy; forgetting; adaptation; per-sample latency	Five-run F1 99.62 $\pm$ 0.04%; forgetting 1.34%; 13.7 ms inference in specified evaluation	Pre-proof and complex deployment assumptions; no held-out episode milestone evaluation.
10	Multi-stage GNN; *Discover Computing*	2026	LANL authentication logs; DARPA Transparent Computing	Temporal interaction graphs and risk-focused attention	Adaptive temporal GNN; propagation-path interpretation	Detection accuracy; noise robustness; runtime	About 5% higher accuracy than the reported GAT comparator	Event/log-centric retrospective detection; no measured lead-time/false-alarm optimisation; SJR Q2/JCR Q3.

Cross-study interpretation. The literature now covers dynamic graphs, strategic attackers, continual learning and cross-dataset testing individually, so their coexistence is not itself new. The proposed study isolates a narrower cause–effect relationship: under the same network traffic generator, capture path, episode budget, and test distribution, *does the scenario-selection policy improve held-out detection and usable pre-milestone warning?*

3. Cross-paper synthesis and refined research gap

Taken together, the evidence does not support claiming to be the first to combine game theory with intrusion detection: Ioannou et al. [2] and Ding et al. [3] already study adversarial adaptation. Nor can early attack forecasting be claimed as new: Nayeri and Rezvani [1] predict future attack interactions. However, the core experiments in [1–3] do not provide a *single controlled network-level evaluation* of strategic versus random scenario-generation policies, with independently timestamped precursor-to-impact milestones, false-alarm constraints and unseen topology/traffic tests. Generalisation findings in [4–6] warn that performance can be dominated by dataset/split artefacts. Graph models [6,8–10] show potential benefit but also sometimes lose to tabular models, implying that graph structure must earn its complexity.

Potential Gap	Forecasting [1]	Adaptive defence [2,3]	Generalisation / graphs [4–10]	Research Opportunity
G1. Causal early-warning lead time	Predicts upcoming alert links from prior alert streams, not separately manifested network milestones.	Optimises evasion/detector resilience but not pre-impact warning coverage.	Temporal graphs are evaluated mainly for classification, shift or noise.	Collect pre-impact stages and valid per-episode T_{lead}, with FP/hour and coverage.
G2. Policy contribution of the data generator	Does not compare scenarios chosen by finite strategic game versus equal-budget random traffic experiments.	Uses RL/evasion loops on benchmark feature spaces.	Model results depend on split and graph design, not a randomised collection intervention.	Hold generator and detector constant; compare fixed, uniform and game strategies.
G3. Network topology / distribution transfer	Alert-edge prediction depends on observed interaction history.	Benchmark perturbations differ from new host-to-host episodes.	Drift/leakage cause strong performance reversals across datasets and graph representations.	Hold out complete scenario families and, where supported, whole topology variants.

Refined gap. Existing work already provides graph-temporal intrusion models, adversarial learning and alert forecasting. What remains insufficiently established is whether *selection of controlled, physically observed network episodes by a small repeated attacker–defender game* improves two independently defined outcomes—current malicious-behaviour detection and early warning before a later measurable network event—without inflating scores through future leakage or overfitting to a specific network. This matters because an operator needs useful warning *time*, not merely a high same-dataset F1. Our study will address this with matched interventions, an independent lab ground-truth timeline and episode-disjoint evaluation. This gap is a *testable hypothesis*, not a claim that no related work exists.

4. Research objectives and hypotheses

Objectives. (1) Record reproducible, time-stamped benign and staged abnormal communication in a segmented virtual network. (2) Train and compare traffic-statistical detectors and, if the observed graph supports it, a compact topology-aware temporal model. (3) Compare fixed, random and strategic scenario selection under equal data-generation budgets. (4) Maximise *valid* early-warning lead time subject to predefined recall and false-alarm constraints. (5) Audit generalisation across unseen scenarios and topology variants.

H1: A strategic scenario selector raises held-out incident recall or PR-AUC against equal-budget fixed and uniform selection. **H2:** It improves the fraction of incidents warned before a milestone and/or the episode-averaged useful lead time at a fixed false-alarm budget. **H3:** Causal temporal and network-relational features add benefit over flow statistics alone on at least some held-out conditions. None is an established result.

5. Proposed research methodology

Step 1. Define the research input

An observation at time t is a *causal* communication multigraph $G_t = (V_t, E_t)$: nodes represent network hosts/services, directed edges represent observed flows within the look-back window $[t - W, t)$, and attributes include protocol, packets, bytes, duration, TCP state and connection timing. The operational traffic task is current benign/suspicious detection at host/window level. Separately, each *incident episode* has independently logged onset $t_{0,e}$ and a later predeclared benign-simulation milestone $t_{m,e}$ (e.g. first controlled inter-segment movement or dummy-data transfer). "Zero-day-like" means a held-out anomalous profile; no actual zero-day exploitation is claimed. An alert's features may use only events visible *before* the alert decision.

Step 2. Data collection

Use Docker Compose with two internal network segments, an explicitly routed capture path, normal client workloads, permitted test clients, web/DNS/SSH-like services and a controlled dummy-data sink. No public-facing vulnerable endpoints, malware, self-propagation or outbound attack traffic. Capture PCAP and Zeek `conn.log` (optionally `dns.log`, `http.log`); Suricata may provide a separate rule-alert comparator but *never authoritative labels*. The scenario orchestrator records topology, seed, service versions, benign load, scenario type and the true onset/milestone times independently of sensors. Pilot a modest suite of ordinary web/API, DNS, approved authentication testing, controlled discovery and staged inter-segment/harmless-transfer episodes. Verify ingress/egress isolation, sensor visibility in both directions and complete log clocks before scaling.

Step 3. Data preprocessing

Use UTC monotonic-aligned capture/manifest timestamps, normalise protocols, flow timeouts and units, resolve duplicate records and non-finite values, and join flows with orchestration events by documented interval/endpoint rules. Preserve manifest IDs for splitting but never feed scenario IDs, absolute IPs or injected action names to the classifier as label shortcuts. Fit imputers/scalers exclusively on training episodes. Keep full raw captures, hashed manifests and explicit excluded-row counts. Build causal windows only after assigning whole episodes to train/validation/test, preventing overlap leakage.

Step 4. Feature engineering and network representation

Flow features: duration, directional packet/byte totals, rates, TCP state and protocol. Causal temporal features: connection frequency, inter-arrival mean/variance/coefficient of variation, burst ratio and rolling deltas. Graph features: distinct neighbours, in/out degree and new inter-segment links; optional graph snapshots aggregate edge attributes. Use a pilot analysis window of $W = 30\,\text{s}$ with sensitivity checks at 15 and 60 s; window length is a preregistered design choice and final selection uses *validation* only. Track missing visibility with masks; do not assume physical routing topology is observable from flow logs alone.

5. Proposed research methodology (continued)

Step 5. Proposed model, scenario game and early warning

Start with a shared causal feature pipeline and two outputs: (a) suspicion probability s_t for current abnormal communication and (b) a sustained binary warning decision $A_t = \mathbb{1}\{s_t \geq \tau\}$, where τ is selected from validation benign/incident episodes. Only if valid prospective labels and sample counts support it, add a separate horizon-risk head $q_t = P(t_m \in (t, t+h] \mid \mathcal{F}_t)$; otherwise detection-generated pre-milestone warnings remain the required early-warning mechanism. A lightweight GraphSAGE/GRU branch is conditional on enough identity-safe graph episodes; a plain temporal model is an explicit comparator.

Define a finite repeated attacker–defender *test scenario game*: the attacker policy chooses among approved traffic-profile/ordering variations $a \in \mathcal{A}$, never executable exploits; the defender policy chooses among predeclared alert/monitoring settings $d \in \mathcal{D}$. Both observe only the last completed episode's *measured* outcomes, not future test labels. Attacker reward is bounded by missed/delayed detection and scenario cost; defender reward credits *valid pre-milestone* warnings and penalises false alarms and response cost. A simple online weighted-regret/EXP3-style selector is sufficient; log the policy probability, available actions, seed and realised rewards. The final classifiers are retrained from separately collected fixed, random and game-selected training corpora. Final assessment uses one frozen independent set common to all three policies.

For incident e, set $t_{a,e}$ to the *first sustained alert* within $[t_{0,e}, t_{m,e})$, where sustained means a validation-fixed persistence rule. The useful lead time is

$$T_{\text{lead},e} = t_{m,e} - t_{a,e} \quad \text{only if } t_{0,e} \leq t_{a,e} < t_{m,e}.$$

A pre-onset alert is a false warning, not artificially large lead time; an episode without a valid alert is *missed*, not omitted from the denominator of overall warning coverage. Optimise *coverage-adjusted* lead time and incident recall while constraining false alarms/hour $\leq \alpha$ on held-out benign episodes. Report conditional lead time separately so a method cannot win by warning on a few easy incidents.

Step 6. Baseline methods

Compare: (i) protocol/flow-statistic threshold rules, (ii) Isolation Forest on normal training traffic, (iii) Random Forest or XGBoost supervised flow-statistic detector, (iv) the same classifier with temporal/graph summary features, and (v) a small GraphSAGE/GRU only if the graph sample size supports it. The scenario-generation baselines are *fixed schedule* and *uniform-random* with the same total lab episode count, duration/traffic budget and scenario action menu as the game policy. Equalise model features, tuning time, final test sets and warning thresholds across policy comparisons.

Step 7. Experimental design

First establish a functioning short-episode pilot and audit timestamp ordering and label coverage. Then run paired experimental blocks under fixed, random and strategic policies with independently varied seeds, benign workloads, attack-profile variations and at least two topology configurations if the lab resources permit. Split complete *episodes* before windowing; no overlapping sessions, hosts/identities when feasible, or scenario-generated duplicates may cross partitions. Hold out one full behavioural family and, for a distinct topology-transfer test, hold out a topology variant. Do not train or tune on final test traces or adapt to their alert outcomes. Pre-register all positive, benign, and pre-milestone inclusion criteria. If a held-out split has no valid future positives, *report forecasting infeasible*, not a fabricated early-warning score.

Step 8. Evaluation metrics

Current detection: macro-F1, positive-class precision/recall, PR-AUC, ROC-AUC when both labels exist, false-alarm rate per benign hour and inference latency. Early warning: pre-milestone warning coverage $N_{\text{warned}}/N_{\text{incident}}$, conditional median and mean T_{lead} among warned episodes, and coverage-adjusted mean $N_{\text{incident}}^{-1} \sum_e T_{\text{lead},e} \mathbb{1}\{\text{valid}_e\}$ (misses contribute zero) with false-alarm/hour constraints. Optional next-stage risk: Brier score and horizon-specific PR-AUC with valid positives. Report per-episode confusion matrices, warning timelines, CPU/RAM cost and group-specific results; never rely on overall accuracy alone.

5. Proposed research methodology (continued)

Step 9. Statistical and comparative analysis

Compute paired method differences on the *same independent test episodes*, grouped by episode/scenario/topology rather than overlapping flow windows. Report 95% bootstrap confidence intervals resampled by episode for incident recall, false alarms/hour, lead-time coverage and the conditional/coverage-adjusted lead-time statistics. If the number of independent repetitions supports it, use a paired permutation or Wilcoxon signed-rank comparison on run-level/episode-level matched outcomes; distinguish exploratory from confirmatory tests. Report both absolute changes and data-generation/training costs.

Step 10. Ablation and component analysis

Ablate (A) strategic selection versus matched uniform random, (B) temporal features removed, (C) graph structure removed while retaining edge statistics, (D) warning persistence/threshold changed on *validation*, and (E) scene/topology transfer versus within-scene tests. Vary window duration, benign traffic mix and held-out behavioural novelty; when supported, evaluate a separate adaptation step without contaminating frozen zero-adaptation results. An ablation that shows no GNN advantage is a legitimate outcome.

Step 11. Error and failure analysis

Inspect concrete false alarms from legitimate bursts, scheduled services or large approved file transfers; missed low-rate or delayed scenarios; detection after the impact milestone; missing bidirectional sensor observations; inconsistent flow termination/timeouts; graph component memorisation; label joins whose timestamps touch window boundaries; and leakage from scenario identifiers. Distinguish (i) early warning, (ii) late but correct detection and (iii) no detection. Plot a representative network communication graph with onset/alert/milestone markings and verify raw PCAP/manifest agreement.

Step 12. Final research output

Publish an emulated topology/Compose specification, safe scenario orchestration, captures or legally distributable derivatives, Zeek configuration, feature dictionary, exact splitting manifests, baseline and selected graph-model code, strategic-policy reward traces, frozen train/validation/test partitions, metrics in CSV/JSON, uncertainty plots, ablation and error tables, and a short operator-facing example showing the suspicious host and *time remaining before the controlled milestone*. Document every data-source limitation, hardware/runtime requirement and failed experiment. Output represents a research prototype, *not* a production intrusion-prevention system.

6. Validity, feasibility and scope

Feasibility. Begin with one small isolated Docker Compose topology, Zeek connection logs, a fixed rule and one tabular detector, and a finite three-policy comparison. Only add a GNN or a more capable defender when the independent observation/label pipeline works. Egress controls, time synchronisation and preservation of raw recordings are mandatory. CUDA is optional; traditional ML and limited graphs can run on CPU. Kubernetes, real self-propagating malware, autonomous exploitation, and an extensive dashboard remain outside scope.

Threats to validity. Controlled traffic may be recognisably unlike enterprise behaviour; scenario timings and IPs may leak labels; a tiny lab may not show genuine topology transfer; class imbalance may hide poor recall; benign periodic/background activity may be underrepresented; an attacker policy may overfit a reference detector. Counter these through benign diversity, fixed budgets, blinding to test outcomes, episode-disjoint splits, a documented holdout and paired uncertainty estimates. A lab generated behavioural variant is not a verified zero-day exploit. A sudden event without observable precursor is not predictably forecastable merely because it happens later.

7. Research methodology architecture

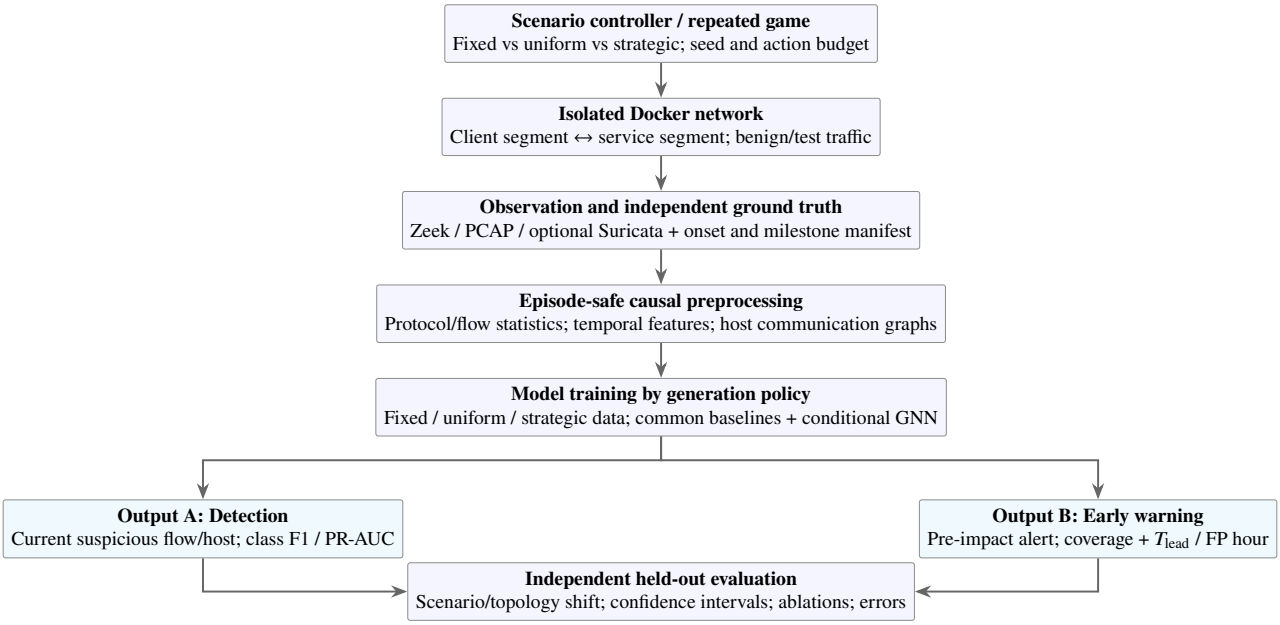


Figure 1. Proposed architecture. Dashed or inferred causal edges are not treated as physical topology. Ground-truth milestones come from the episode controller, *not* an IDS alert. Policy learning and model tuning never see final test episode outcomes. Response/containment recommendations and a next-stage prediction head are optional extensions.

8. Intended contribution

The intended contribution is a *controlled evaluation framework*, not an unsupported claim to a new universal IDS: a network-level test of whether strategic scenario selection supplies better training evidence than fixed/random generation, coupled with a precise early-warning protocol. The research can succeed as a critical comparison even if the simplest statistical detector matches the GNN or the game policy provides no benefit. No local results have yet been collected.

References (additional papers; continued on next page)

[1] Z. Makki Nayeri and M. Rezvani, "Alert prediction in computer networks using transformer-based temporal graph neural networks: Identifying the next victim," *Journal of Network and Computer Applications*, vol. 249, art. 104455, 2026. DOI 10.1016/j.jnca.2026.104455.

[2] I. Ioannou et al., "Adaptive active-defense hardening of ML-based NIDS against RL-driven adversaries: A comparative analysis with static defenses," *Journal of Information Security and Applications*, vol. 100, art. 104496, 2026. DOI 10.1016/j.jisa.2026.104496.

[3] J. Ding, Z. Shen, and W. Liu, "Game-Theoretic Cost-Sensitive Adversarial Training for Robust Cloud Intrusion Detection Against GAN-Based Evasion Attacks," *Applied Sciences*, vol. 16, art. 3944, 2026. DOI 10.3390/app16083944.

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[4] K. K. Eren et al., "Distributional Drift in IoT Intrusion Detection Systems: Implications for Cross-Dataset Generalisation," *Electronics*, vol. 15, art. 2307, 2026. DOI 10.3390/electronics15112307.

[5] S. Mishra, N. S. Alshammari, H. Hussain, and R. A. Alfahidah, "A cross-dataset harmonized intrusion detection framework with statistically validated multi-model learning," *PLOS ONE*, vol. 21, e0346982, 2026. DOI 10.1371/journal.pone.0346982.

[6] S. R. Nune et al., "Hierarchical graph attention and memory-augmented learning for spatio-temporal network intrusion detection," *Scientific Reports*, article in press, 2026. DOI 10.1038/s41598-026-78200-2.

[7] T. Wang, Q. Liu, H. Shen, and G. Bai, "IDNet: Instance-adaptive dynamic network with adversarial training for intrusion detection," *Computer Networks*, vol. 283, art. 112336, 2026. DOI 10.1016/j.comnet.2026.112336.

[8] J. Yao and B. Koirala, "Defense-Aware Temporal Graph Neural Network for fault-tolerant intrusion detection in IIoT–5G environments," *Cluster Computing*, vol. 29, art. 319, 2026. DOI 10.1007/s10586-026-06037-5.

[9] M. Javed et al., "HECATE-IDS: Explainable continual temporal graph learning for adaptive intrusion detection in IoT networks," *Pervasive and Mobile Computing*, accepted pre-proof, 2026. DOI 10.1016/j.pmcj.2026.102302.

[10] T. N. Suharsono, T. B. Kurniawan, D. A. Dewi, and H. M. Kurniawan, "Adaptive temporal graph neural networks for detecting coordinated multi-stage cyber-attacks in enterprise systems," *Discover Computing*, vol. 29, art. 455, 2026. DOI 10.1007/s10791-026-10347-y.

Phase 1 background (not counted among the ten additional studies): [P1] Deepanramkumar and Sharmila, 10.1007/s44443-026-01260-x; [P2] Bala et al., 10.1016/j.dt.2026.06.002; [P3] Afrifah et al., 10.1007/s10207-026-01340-z.

Reproducibility and eligibility statement

Journal eligibility. Verified 2025 journal rankings for the focal studies are Q1. Additional venues include journals classified Q1/Q2 in at least one 2025 SJR or JCR category: for example, *Applied Sciences* and *Electronics* list JCR Q2; *Pervasive and Mobile Computing* SJR Q1/JCR Q2; *Discover Computing* SJR Q2/JCR Q3. The departmental choice of indexing system and category governs final acceptance. Rankings were compared with publisher and ISSN-linked journal-index profiles; they are not claimed to be 2026 quartiles.

Public datasets and code (selected). Phase 1 supplies public project repositories for GraphINR-IDS (https://github.com/Deepan777/GraphINR-IDS) and TF-GATv2 (https://github.com/jencyyyyy/TF-GATv2), with author-specified original benchmark datasets. Afrifah et al. give https://github.com/AfrifaW/Multi-Stage-Attack-Detection, but access/run success has not been independently confirmed. Of the additional papers, Nayeri and Rezvani list https://github.com/ZahraMakkiNayeri/TGNE-TA.git in the article, Mishra et al. specify https://github.com/rubaamff/CrossHIDS, Nune et al. cite https://doi.org/10.5281/zenodo.23293134, and Yao and Koirala cite https://doi.org/10.5281/zenodo.18550299. These author-disclosed links are leads for reproduction, *not* proof that all code, data versions, models or environments have been independently downloaded or executed. Ioannou et al. offer data upon request; the source PDFs for several other works do not establish an accessible official code repository. No experiment was executed for this literature review.

Dataset integrity and publication status. UNSW-NB15, CIC-IDS2017, ToN-IoT and Edge-IIoTset have different schemas, labelling regimes and temporal completeness; they cannot be silently pooled. HECATE-IDS is a 2026 accepted journal pre-proof rather than final typesetting, and STAR-IDS is marked "article in press." The paper-specific metrics remain attributed to their authors. All implementation/sample counts in this proposal are plans only.